

Problem-Solution Fit

Date	03-07-2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Primary: Small & marginal farmers in AP/TS with 1-5 acre land Secondary: FPOs, Agriculture officers, Agri-input dealers Tertiary: Govt policymakers for subsidy planning	2. PROBLEMS / PAINS + ITS FREQUENCY [PR] P1: Wrong crop selection — 30-40% income loss, Every Season P2: Market price crash due to glut — ₹25k/acre loss, 2 times/year P3: No weather alerts — Crop failure, Once in 3 years P4: Information scattered — 5+ hours/week waste	3. TRIGGERS TO ACT [TR] T1: Monsoon onset — Need to decide crop in 7 days T2: Previous season loss — "this time must not repeat the mistake" mindset T3: Neighbor got good price — FOMO T4: Govt subsidy announcement for specific crop
4. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Budget: Can't afford ₹5000+ paid apps Devices: 60% use feature phones, not smartphones Literacy: 40% can't read English, low digital skills Network: Poor internet in villages	7. BEHAVIOR + ITS INTENSITY [BE] High Intensity: Call agriculture officer/AEO for advice - 80% farmers do this weekly Medium: Watch TV programs like Annadata - 60% farmers High: Ask	8. CHANNELS OF BEHAVIOR [CH] ONLINE WhatsApp Groups: village farmer groups share the OptiCrop link GitHub Repository: open-source access for developers & FPOs to self-host OFFLINE Krishi Seva Kendra: demo booths set up by FPOs Agriculture Officer Meetings: village-level monthly meetings

	neighbor farmers what they planted - 90% farmers Result: Herd behavior, market glut	for introduction
5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros: Kisan Suvidha app - Free, govt backed Cons: No ML prediction, generic advice, English only Pros: Private apps like Plantix - Good pest detection Cons: ₹2000/year cost, no market price integration	9. PROBLEM ROOT / CAUSE [RC] R1: Decisions based on tradition/neighbors, not data R2: No single platform combining soil + weather + market R3: Existing solutions not vernacular + not SMS friendly R4: Illiteracy & tech phobia	10. YOUR SOLUTION [SL] OptiCrop ML Engine: Voice/Text Input: farmer enters soil & climate data (N, P, K, temperature, humidity, ph, rainfall) Instant Crop Recommendation: Logistic Regression model trained on 2,200+ records, 97.3% test accuracy, 22 crop classes Simple Web Interface: Flask app with Home, About, Find-Your-Crop, and Result pages — accessible from any browser Free & Open Source: Hosted on GitHub (github.com/pavanvoleti/OptiCrop), zero cost to the farmer
6. EMOTIONS BEFORE / AFTER [EM] BEFORE — Anxiety: "Will the rains fail and ruin the crop? How will I repay the loan?" AFTER — Confidence: "The ML model predicted with 97.3% accuracy. If I plant this crop, it should be profitable."		